

Engineering Technology Department

ET575 - Introduction to C++ Programming Design and Implementation

Exam 3 - review

Question 1) Given the following C++ code:

```
#include<iostream>
#include<cmath>
using namespace std;
const double PI = 3.14159;

double collect_diameter(){
    double d;
    cout<<"Enter a diameter: ";
    cin>> d;
    return d;
}
int totalPrice(double diameter, double price){           // diameter = 8      price = 2.5
    double a, r;
    r = diameter/static_cast<double>(2);
    a = PI*pow(r,2);
    return ((a*price)+2);                               // return = 127.66 ~= 127
}
void testing(double d,double c){                         // d = 8              c = 2.5
    double total;
    char ans;
    do{
        total = totalPrice(d,c);
        cout<<"The total cost is $ "<<total<<endl;
        cout<<"Test again? (y/n) ";
        cin>>ans;
        cout<<endl;
    }while(ans=='Y' || ans =='y');
}
int main(){
    double dia = collect_diameter(), cost = 2.5;
    testing(dia,cost);
    return 0;
}
```

When the program is run, the user types and enters 8, what should be prompting in the terminal?

Ans: The Total Cost is \$ 127

Question 2) pointers and reference

Given the following functions, fill-in the blank of the main() function after the program is run.

```
void printvalue(string v){
    cout<<"Pass value = "<<v<<endl;
}
void passref(string& v){
    cout<<"Pass refernce = "<<v<<endl;
    v = "Good Morning";
}
void passmemory(string* v){
    cout<<"Pass refernce = "<<v<<endl;
}

int main(){
    string something = "ET575";
    printvalue(something);    // result in Terminal = Pass value = ET575
    passref(something);       // result in Terminal = Pass reference = ET575
    printvalue(something);    // result in Terminal = Pass value = Good Morning
    passmemory(&something);   // result in Terminal = Pass reference =
    <memories address>
    return 0;
}
```

Question 3) complete the C++ code below where function `functionArray()` will read all numbers in array and return the sum of all even numbers.

```
#include <iostream>
using namespace std;
int functionArray(int* arr, int size){
    int sum=0;
    for (int i = 0; i < size; i++){
        if (arr[i] % 2 == 0)
            sum += arr [i];
    }
    return sum;
}
```

